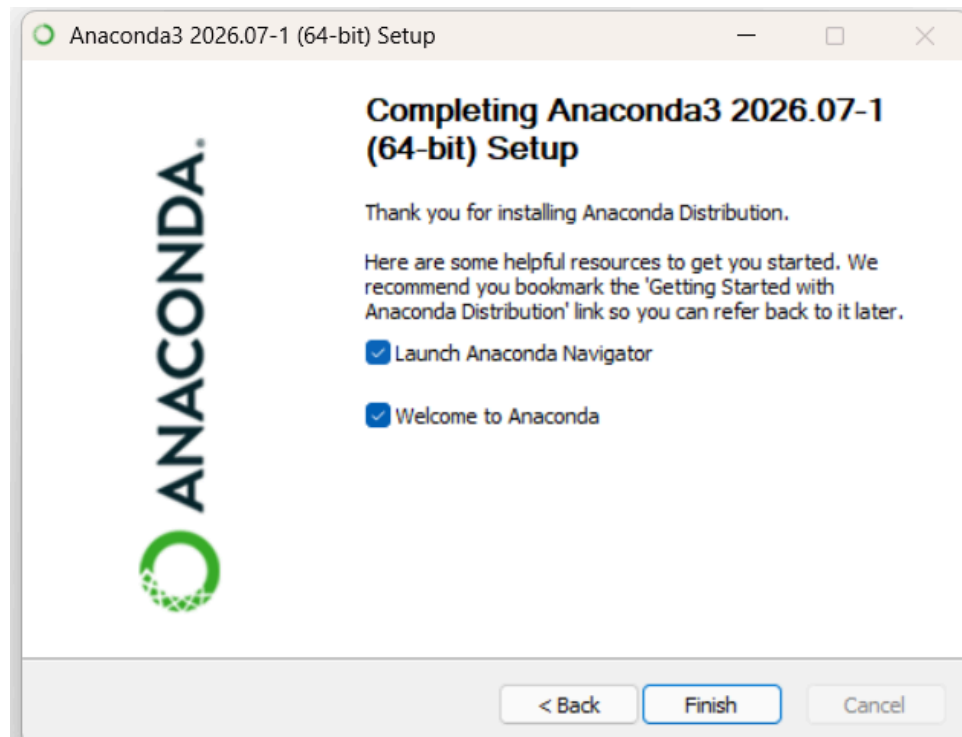
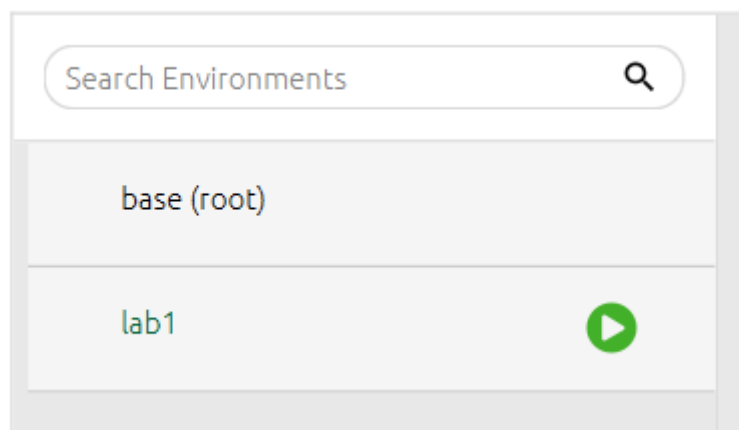
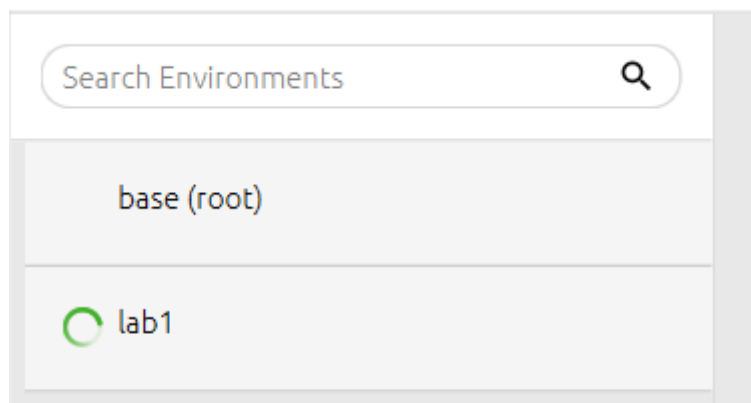


Часть 1. Работа в среде Anaconda

1. Установка среды



2. Создание и активация изолированного виртуального окружения через графический интерфейс:



Через терминал:

```
(base) C:\Users\Admin>conda create -n lab1_x2 python=3.14.7
3 channel Terms of Service accepted
Retrieving notices: done
Channels:
- defaults
Platform: win-64
Collecting package metadata (repodata.json): done
Solving environment: done

## Package Plan ##

environment location: D:\Anaconda_environments_uni\lab1_x2

added / updated specs:
- python=3.14.7
```

```
(base) C:\Users\Admin>conda activate lab1_x2

(D:\Anaconda_environments_uni\lab1_x2) C:\Users\Admin>|
```

3. Установленные пакеты (терминал):

```
(base) C:\Users\Admin>conda list numpy
# packages in environment at C:\Users\Admin\anaconda3:
#
# Name                      Version      Build                Channel
numpy                       2.4.6        py314h650627e_0
numpy-base                  2.4.6        py314h5fb32f3_0
numpydoc                    1.10.0       py314haa95532_0
```

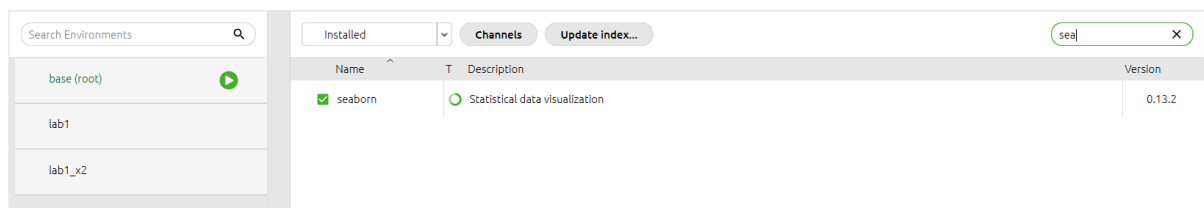
```
(base) C:\Users\Admin>conda list pandas
# packages in environment at C:\Users\Admin\anaconda3:
#
# Name                      Version      Build                Channel
pandas                      3.0.3        py314hf1216d0_0
```

```
(base) C:\Users\Admin>conda list sklearn
# packages in environment at C:\Users\Admin\anaconda3:
#
# Name                      Version      Build                Channel
sklearn-compat              0.1.6        py314haa95532_0
```

```
(base) C:\Users\Admin>conda list seaborn
# packages in environment at C:\Users\Admin\anaconda3:
#
# Name                      Version      Build                Channel
seaborn                     0.13.2       py314haa95532_3
```

```
(base) C:\Users\Admin>conda list matplotlib
# packages in environment at C:\Users\Admin\anaconda3:
#
# Name                      Version      Build                Channel
matplotlib                  3.11.0       py314haa95532_0
matplotlib-base              3.11.0       py314hce83770_0
matplotlib-inline            0.2.2        py314haa95532_0
```

Проверка установленных пакетов через графический интерфейс:



Установка:

☒	seaborn	Statistical data visualization	0.13.2
☐	the_silver_searcher	A code searching tool similar to ack, with a focus on speed.	2.2.0
☐	tokenizers	Fast state-of-the-art tokenizers optimized for research and production	0.23.1
☐	whoosh	Whoosh is a fast, featureful full-text indexing and searching library implemented in pure python	2.7.4

44 packages available matching "sea" 1 package selected

Apply Clear

Программное определение версий пакетов:

```

Определим установленные версии пакетов:

[6]: import numpy as np
import pandas as pd
import matplotlib as mpl
import sklearn
import seaborn as sns

packages = {
    'numpy': np.__version__,
    'pandas': pd.__version__,
    'matplotlib': mpl.__version__,
    'scikit-learn': sklearn.__version__,
    'seaborn': sns.__version__,
}

Выведем:

[7]: for pkg, version in packages.items():
print(f'{pkg:15} : {version}')

numpy          : 2.5.2
pandas         : 3.0.5
matplotlib     : 3.11.0
scikit-learn   : 1.9.0
seaborn        : 0.13.2

```

Все установленные пакеты в данном окружении через терминал:

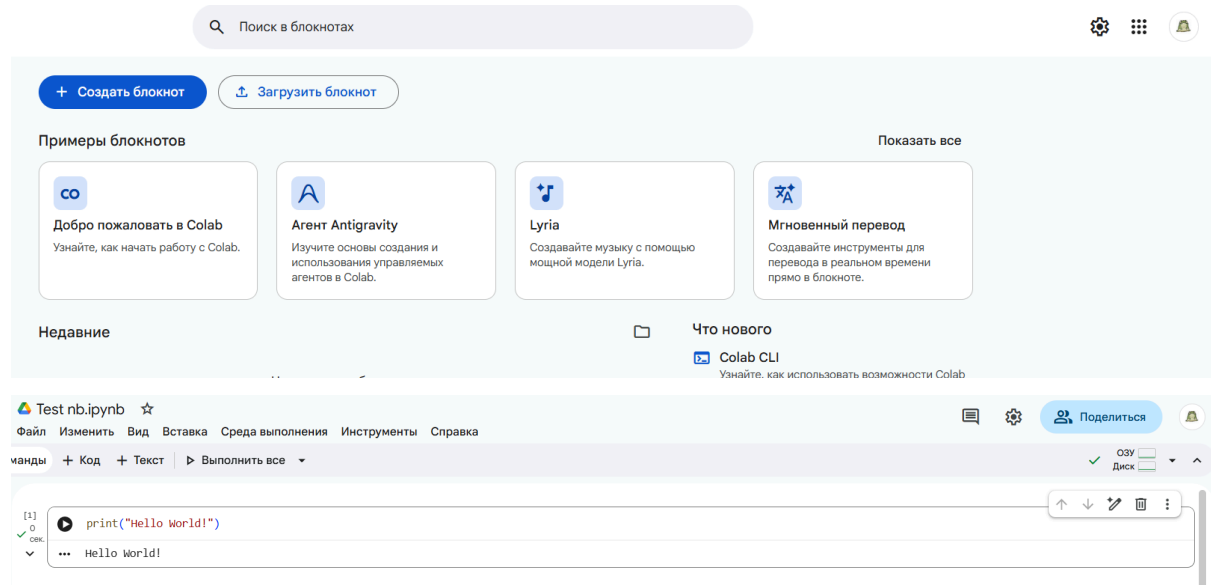
```

(D:\Anaconda_environments_uni\lab1_x2) C:\Users\Admin>conda list
# packages in environment at D:\Anaconda_environments_uni\lab1_x2:
#
# Name                      Version           Build                Channel
bz2                        1.0.8             h2bfff1b_6           conda-forge
ca-certificates           2026.8.13         haa95532_0           conda-forge
libexpat                  2.8.4             hd7fb8db_0           conda-forge
libffi                    3.4.8             h42d73b9_3           conda-forge
libmpdec                  4.0.1             h1c6eee0_0           conda-forge
libzlib                   1.3.2             h1c6eee0_0           conda-forge
lz4-c                     1.9.4             h1c6eee0_5           conda-forge
openssl                   3.5.8             hbb43b14_0           conda-forge
pip                       26.2.1            pyh0d26453_0         conda-forge
python                    3.14.7            h7ce57fb_101_cp314   conda-forge
python_abi                3.14              4_cp314              conda-forge
sqlite                    3.53.2            hee5a0db_0           conda-forge
tk                        9.0.4             hd2208ad_1           conda-forge
tzdata                    2026c             he532380_0           conda-forge
ucrt                      10.0.22621.0     haa95532_0           conda-forge
vc                        14.3              h2df5915_12          conda-forge
vc14_runtime              14.44.35208       h4927774_12          conda-forge
vs2015_runtime            14.44.35208       ha6b5a95_12          conda-forge
xz                        5.8.2             h53af0af_0           conda-forge
zlib                      1.3.2             h1c6eee0_0           conda-forge
zstd                      1.5.7             h56299aa_0           conda-forge

```

Часть 2. Работа в среде Google Colab

1. Авторизация в среде и создание блокнота



2. Оценка наличия пакетов

!pip list | grep sklearn

```
... sklearn-compat      0.1.6
    sklearn-pandas      2.2.0
```

!pip list | grep matplotlib

```
... matplotlib          3.10.0
    matplotlib-inline    0.2.2
    matplotlib-venn      1.1.2
```

!pip list | grep numpy

```
... numpy                2.1.3
```

!pip list | grep pandas

```
... geopandas           1.1.4
    pandas               2.2.3
    pandas-datareader     0.11.1
    pandas-gbq            0.30.0
    pandas-stubs          2.2.3.250527
    sklearn-pandas        2.2.0
```

!pip list | grep seaborn

```
... seaborn              0.13.2
```

Программное определение версий пакетов:

```

import numpy as np
import pandas as pd
import matplotlib as mpl
import sklearn
import seaborn as sns

packages = {
    'numpy': np.__version__,
    'pandas': pd.__version__,
    'matplotlib': mpl.__version__,
    'scikit-learn': sklearn.__version__,
    'seaborn': sns.__version__,
}

```

Вот такие вот:

```

▶ for pkg, version in packages.items():
    print(f"{pkg:15} : {version}")

```

```

... numpy          : 2.1.3
... pandas         : 2.2.3
... matplotlib     : 3.10.0
... scikit-learn   : 1.6.1
... seaborn        : 0.13.2

```

3. Переключение между режимами работы среды:

Сменить среду выполнения

Тип среды выполнения

Python 3 ▼

Аппаратный ускоритель ?

- ☒ ЦП
 ☐ Графический процессор T4
- ☐ Графический процессор H100
- ☐ Графический процессор G4
- ☐ Графический процессор A100
- ☐ Графический процессор L4
 ☐ TPU v5e-1
- ☐ TPU v6e-1

ⓘ Нужен доступ к мощным графическим процессорам?
[Перейти на Colab Pro+](#)

```
[5] ltpu-info
✓ 12
cek. Libtpu version: 0.0.21.1
Accelerator type: v5e

TPU Chips
```

Chip	Type	Devices	PID
/dev/vfio/0	TPU v5e chip	1	None

```
WARNING: Libtpu metrics unavailable. Is there a framework using the TPU? See
```

Часть 3. Экспериментальная проверка на наборах данных

```
[18]: import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

[19]: df = pd.read_csv('student_performance_dataset.csv')
df.head(10)

[19]: student_id  gender  study_time_hours  attendance_percent  sleep_hours  parental_education  internet_access  extracurricular_activities  part_time_job  previous_grade
0           1    Male             4.0             98.0           6.5         Bachelors             Yes              Yes              No             76.9
1           2   Female             6.3            100.0           5.7         High School          Yes              Yes              Yes             75.5
2           3    Male             4.9             85.3           7.9         Bachelors             Yes              No              Yes             88.5
3           4    Male             2.6             77.5           8.0           NaN              Yes              Yes              No             85.1
4           5    Male             2.2             89.6           4.6         Bachelors             Yes              No              Yes             61.8
5           6   Female             4.2             78.2           5.7         High School          Yes              No              No             79.8
6           7    Male             1.5            100.0           5.0         High School          Yes              Yes              Yes             58.6
7           8    Male             6.2             86.4           6.0         High School          Yes              Yes              No             78.4
8           9    Male             5.3             81.3           6.7         Bachelors             Yes              No              No             63.8
9          10   Female             2.8             86.8           5.1         Bachelors             Yes              Yes              No             57.7

[20]: gender_counts = df['gender'].value_counts()
print(gender_counts)

gender
Female    510
Male      490
Name: count, dtype: int64

[21]: print(df.columns.tolist())

['student_id', 'gender', 'study_time_hours', 'attendance_percent', 'sleep_hours', 'parental_education', 'internet_access', 'extracurricular_activities', 'part_time_job', 'previous_grade', 'final_exam_score', 'final_grade']

[22]: plt.figure(figsize=(8, 6))
sns.countplot(data=df, x='gender')

plt.title('Gender distribution among students', fontsize=14)
plt.xlabel('Gender', fontsize=12)
plt.ylabel('Students', fontsize=12)
plt.show()
```

Команды + Код + Текст ▶ Выполнить все ▾

Подключение библиотек и загрузка датасета:

[30] 0 сек. import matplotlib.pyplot as plt
import pandas as pd
import seaborn as sb

[31] 0 сек. df = pd.read_csv('imdb_top_1000.csv')
df.head(10)

1 to 10 of 10 entries

index	Poster_Link	Series_Title	Release_Year
0	https://m.media-amazon.com/images/M/MV5BMDFKYtC0MGEtZmNhMC00ZDIzLWFmNTI0MDE1ZmRlYWMwMWFmXkEyXkFqcGdeQXVyMTMxODk2OTU@._V1_UX67_CR0,0,67,98_AL_.jpg	The Shawshank Redemption	1994
1	https://m.media-amazon.com/images/M/MV5BM2MyNjYxNmUyTAWNi00MTYxLWJmNWY1YzZlODY3ZTk3OTFkXkEyXkFqcGdeQXVyNzkwMjQ5NzZlM@._V1_UY98_CR1,0,67,98_AL_.jpg	The Godfather	1972
2	https://m.media-amazon.com/images/M/MV5BMTMxNTMwODM0NF5BMi5BanBnXkFtZTcwODAyMTk2Mw@@._V1_UX67_CR0,0,67,98_AL_.jpg	The Dark Knight	2008

Переменные Терминал 00